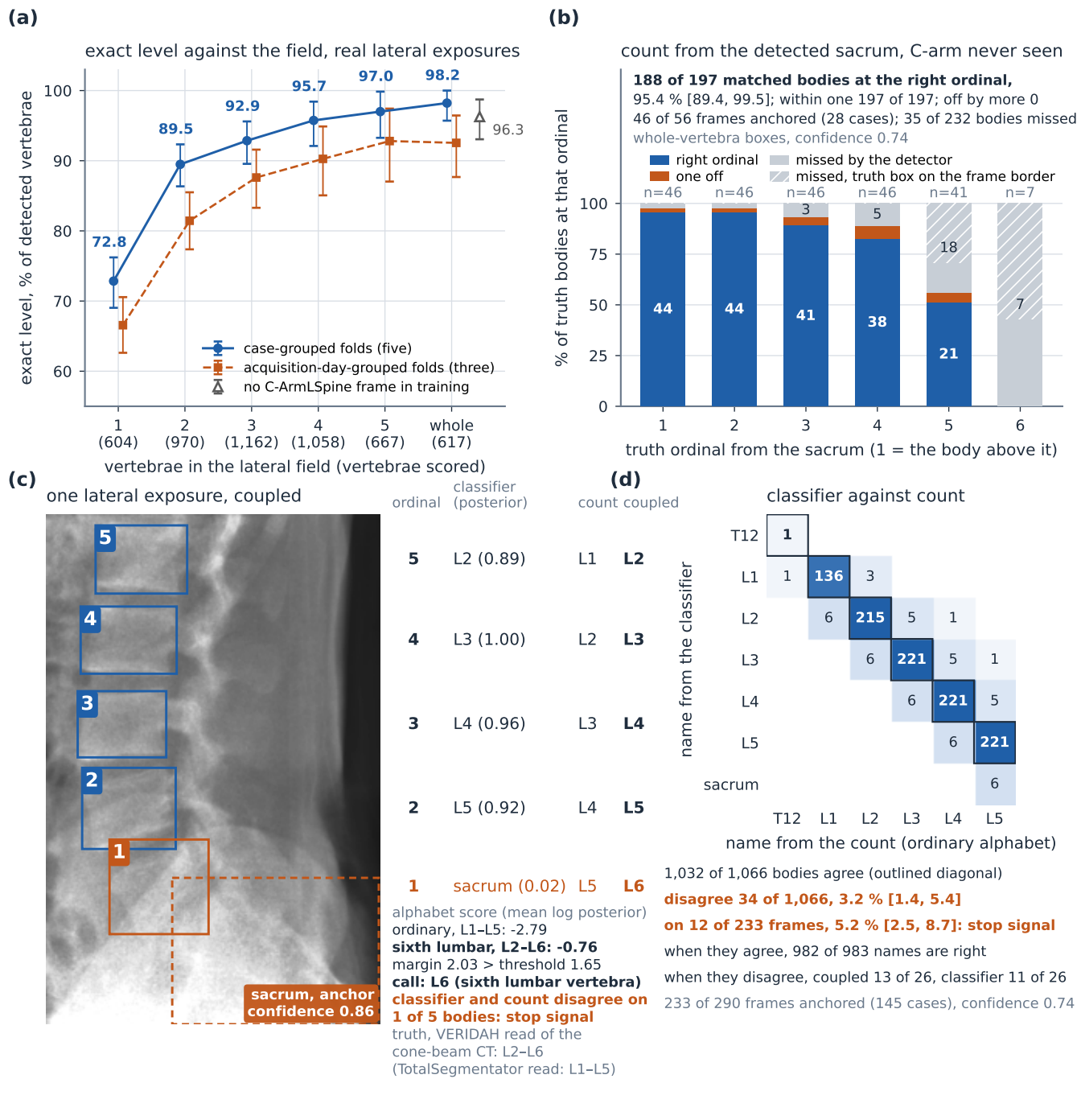

(a)
exact level against the field, real lateral exposures
exact level, % of detected vertebrae
100
90
80
70
60
72.8
89.5
92.9
95.7
97.0
98.2
96.3
case-grouped folds (five)
acquisition-day-grouped folds (three)
no C-ArmLSpine frame in training
1 (604)
2 (970)
3 (1,162)
4 (1,058)
5 (667)
whole (617)
vertebrae in the lateral field (vertebrae scored)
(b)
count from the detected sacrum, C-arm never seen
188 of 197 matched bodies at the right ordinal,
95.4 % [89.4, 99.5]; within one 197 of 197; off by more 0
46 of 56 frames anchored (28 cases); 35 of 232 bodies missed
whole-vertebra boxes, confidence 0.74
right ordinal
one off
missed by the detector
missed, truth box on the frame border
n=46
n=46
n=46
n=46
n=41
n=7
% of truth bodies at that ordinal
100
75
50
25
0
44
44
41
3
38
5
21
18
7
1
2
3
4
5
6
truth ordinal from the sacrum (1 = the body above it)
(c)
one lateral exposure, coupled
5
4
3
2
1
sacrum, anchor confidence 0.86
ordinal
classifier (posterior)
count coupled
5 L2 (0.89) L1 L2
4 L3 (1.00) L2 L3
3 L4 (0.96) L3 L4
2 L5 (0.92) L4 L5
1 sacrum (0.02) L5 L6
alphabet score (mean log posterior)
ordinary, L1–L5: -2.79
sixth lumbar, L2–L6: -0.76
margin 2.03 > threshold 1.65
call: L6 (sixth lumbar vertebra)
classifier and count disagree on 1 of 5 bodies: stop signal
truth, VERIDAH read of the cone-beam CT: L2–L6
(TotalSegmentator read: L1–L5)
(d)
classifier against count
name from the classifier
T12
L1
L2
L3
L4
L5
sacrum
1
1 136 3
6 215 5 1
6 221 5 1
6 221 5
6 221
6
T12 L1 L2 L3 L4 L5
name from the count (ordinary alphabet)
1,032 of 1,066 bodies agree (outlined diagonal)
disagree 34 of 1,066, 3.2 % [1.4, 5.4]
on 12 of 233 frames, 5.2 % [2.5, 8.7]: stop signal
when they agree, 982 of 983 names are right
when they disagree, coupled 13 of 26, classifier 11 of 26
233 of 290 frames anchored (145 cases), confidence 0.74